

SciCLIP: Parameter-Efficient Fine-Tuning of CLIP for Cross-Modal Scientific Figure Retrieval

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Abstract

Retrieving scientific figures (plots, diagrams, and charts) from natural language queries is a cross-modal retrieval task with limited prior work. General-domain vision-language models such as CLIP perform well on natural images but show a clear domain gap when applied to academic figures with specialized visual and textual semantics. In this paper, we describe **SciCLIP**, a lightweight adaptation method that applies Low-Rank Adaptation (LoRA) to a frozen clip-vit-base-patch32 backbone, targeting the query, key, and value projection layers of both encoders.

We train on an 18,000-sample subset of the SciCap dataset and evaluate on a 2,000-candidate retrieval pool. The adapted model improves Median Rank from 110.0 to 34.0 and MRR from 0.1564 to 0.2324 relative to the zero-shot baseline. While absolute retrieval accuracy remains moderate, these results show that even minimal adaptation ($\sim 0.5\%$ trainable parameters) narrows the domain gap. We also present an ablation over LoRA ranks ($r \in \{4, 8, 16\}$), observing that lower ranks have a regularizing effect under limited training data, and describe a Gradio-based interactive retrieval demo backed by a FAISS index.

1 Introduction

Cross-modal information retrieval—searching for visual content using natural language queries—has advanced considerably through joint vision-language representation learning. In academic contexts, scientific figures often summarize key results of research papers, and the ability to retrieve them via textual queries has applications in literature review, survey generation, and academic search.

However, scientific figure retrieval poses distinct challenges. Models like CLIP (Radford et al., 2021), trained on large-scale natural image-text pairs, lack exposure to the specialized vocabulary and visual conventions of academic diagrams.

Terms such as “BLEU score ablation” or “t-SNE embedding scatter plot” have little representation in general web corpora, and the visual structure of scientific figures (axes, legends, schematic boxes) differs substantially from natural photographs.

Full fine-tuning is expensive and can degrade the model’s original representations. Parameter-Efficient Fine-Tuning (PEFT) methods offer an alternative. In particular, Low-Rank Adaptation (LoRA) (Hu et al., 2022) injects small trainable matrices into frozen transformer layers, allowing domain adaptation with minimal parameter overhead.

In this work, we apply LoRA-based adaptation to CLIP for scientific figure retrieval and report the following contributions:

- **Data Pipeline:** We describe a data configuration and preprocessing pipeline for the CrowdAIIab/scicap dataset, including structural caption index removal and length filtering.
- **LoRA-Adapted Retrieval:** We show that adapting $\sim 0.5\%$ of CLIP’s parameters with LoRA produces consistent improvements over the zero-shot baseline across all measured retrieval metrics.
- **Rank Ablation:** We compare LoRA ranks $r \in \{4, 8, 16\}$ and find that lower ranks tend to generalize better under our limited training regime, while higher ranks show signs of diminishing returns.
- **Interactive Demo:** We provide a Gradio-based side-by-side comparison interface using FAISS for efficient nearest-neighbor search.

2 Related Work

2.1 Vision-Language Pretraining and CLIP

CLIP (Radford et al., 2021) learns joint vision-language representations through contrastive pre-training on approximately 400 million image-text pairs. Its twin-tower architecture projects images and text into a shared embedding space, enabling zero-shot transfer to a range of downstream tasks. However, because its training data consists predominantly of natural images with general web text, performance degrades on domain-specific content such as scientific figures.

2.2 Parameter-Efficient Fine-Tuning and LoRA

LoRA (Hu et al., 2022) adapts pre-trained models by decomposing weight updates into low-rank matrices. For a frozen weight matrix $W_0 \in R^{d \times k}$, the adapted weight is:

$$W = W_0 + \frac{\alpha}{r} B \cdot A \quad (1)$$

where $B \in R^{d \times r}$, $A \in R^{r \times k}$, and $r \ll \min(d, k)$. This formulation has been applied to language models and diffusion models; to our knowledge, its use for adapting vision-language models to specialized retrieval domains has received less attention.

2.3 Scientific Figure Datasets

The SciCap dataset (Hsu et al., 2021) is a large-scale collection of over 400,000 scientific figures paired with their captions. Due to its massive scale, full-dataset fine-tuning is computationally prohibitive for constrained environments. We describe our approach to sampling and preprocessing a representative subset in Section 3.2.

3 Methodology

3.1 Model Architecture and Adapter Placement

We use openai/clip-vit-base-patch32 as the backbone. The visual encoder (ViT-B/32) maps 224×224 images to 512-dimensional vectors; the text encoder maps token sequences (max length 77) to vectors of the same dimensionality.

LoRA adapters are injected into the query, key, and value projection layers (q_proj, k_proj, v_proj) of both the visual and textual attention blocks. All original weights remain frozen. At rank $r = 8$, the trainable parameters constitute approximately 0.5% of the total model parameters.

3.2 Dataset Configuration and Preprocessing

We utilize the CrowdAIIab/scicap dataset, a large-scale collection of over 416,000 scientific figures extracted from arXiv computer science papers. Due to computational constraints, we sample a subset of 20,000 image-caption pairs for our experiments. Table 1 summarizes the dataset splits.

Configuration	Count / Detail
Source Dataset	SciCap (arXiv)
Total Available Figures	~416,000
Sampled Subset	20,000
Training Split	18,000
Evaluation Split	2,000

Table 1: Dataset configuration. We use a 20,000-sample subset of the full SciCap dataset for training and evaluation.

Annotations are loaded from train.json in COCO format, and image IDs are mapped to file names. We apply the following preprocessing steps to ensure high-quality training pairs:

- **Caption Index Removal:** Captions frequently begin with structural prefixes (e.g., “Figure 3:” or “Fig. 4(b):”) that can serve as superficial shortcuts during training. We use the caption_no_index field to strip these prefixes, encouraging the model to attend to semantic content.
- **Length Filtering:** We discard captions shorter than 20 characters or longer than 300 characters. Short captions lack informative content; long captions exceed CLIP’s 77-token input limit.

3.3 Training Objective

We train with the symmetric InfoNCE contrastive loss (van den Oord et al., 2018). For a batch of N image-text pairs with normalized embeddings $\{v_i\}_{i=1}^N$ and $\{t_i\}_{i=1}^N$, the pairwise similarity is:

$$S_{i,j} = e^{\tau \cdot (v_i^\top t_j)} \quad (2)$$

where τ is a learned temperature. The loss averages the image-to-text and text-to-image directions:

$$\mathcal{L}_{v2t} = -\frac{1}{N} \sum_{i=1}^N \log \frac{S_{i,i}}{\sum_{j=1}^N S_{i,j}} \quad (3)$$

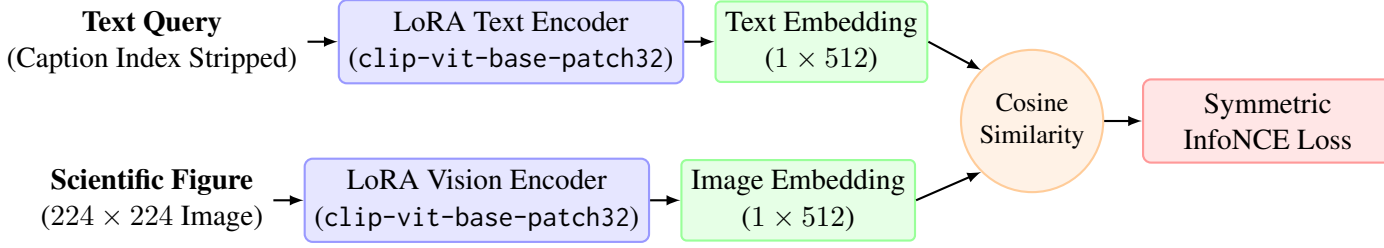


Figure 1: SciCLIP training pipeline. LoRA adapters are applied to the query, key, and value projection layers of both encoders. The model is trained with a symmetric InfoNCE loss over paired scientific figures and captions.

$$\mathcal{L}_{t2v} = -\frac{1}{N} \sum_{i=1}^N \log \frac{S_{i,i}}{\sum_{j=1}^N S_{j,i}} \quad (4)$$

$$\mathcal{L}_{total} = \frac{\mathcal{L}_{v2t} + \mathcal{L}_{t2v}}{2} \quad (5)$$

This objective encourages the model to pull the embeddings of matching image–text pairs closer together in the joint vector space while pushing apart the embeddings of mismatched pairs. By optimizing this loss, the model learns to effectively align the visual features of abstract scientific figures with their corresponding technical descriptions.

4 Experiments

4.1 Setup

Training is conducted on an NVIDIA A100 GPU via Google Colab. From 20,000 preprocessed samples, we use 18,000 for training and 2,000 for evaluation (90/10 split). We train for 5 epochs with batch size 64, using the AdamW optimizer (learning rate 1×10^{-4} , weight decay 0.01) with a cosine annealing schedule.

4.2 Metrics

We evaluate using standard retrieval metrics on the 2,000-candidate pool:

- **Recall@k:** Fraction of queries for which the correct image appears in the top k results ($k \in \{1, 5, 10\}$).
- **Mean Reciprocal Rank (MRR):** $MRR = \frac{1}{M} \sum_{i=1}^M \frac{1}{\text{rank}_i}$
- **Median Rank:** Median position of the correct image across all queries (lower is better).

4.3 Results

Table 2 summarizes the retrieval performance. All three LoRA configurations outperform the zero-shot baseline across every metric, with the best variant ($r = 8$) improving Recall@5 from 19.35%

to 29.60% and reducing Median Rank from 110 to 34. Absolute recall remains moderate, indicating that the task is far from solved.

4.4 Analysis

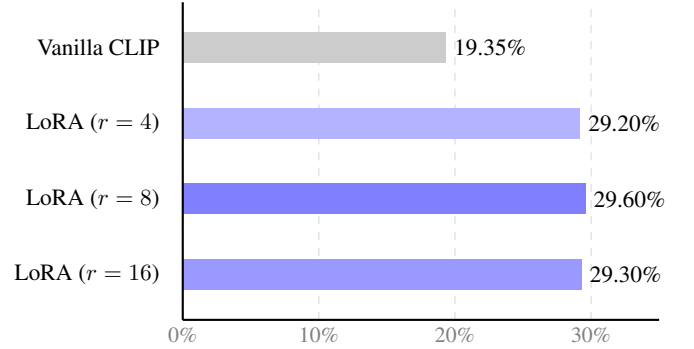


Figure 2: Recall@5 by model configuration. LoRA adaptation consistently improves over the baseline; differences among rank settings are small.

We observe the following patterns:

1. **Baseline improvement:** All three LoRA configurations improve over the zero-shot baseline. Recall@5 increases from 19.35% to roughly 29–30%, and the Median Rank drops from 110 to 34–36. These relative improvements are clear, but the absolute recall values (e.g., R@1 of 16–17%) indicate that the task remains challenging and adaptation alone does not close the domain gap.
2. **Effect of rank on generalization:** The smallest rank ($r = 4$) achieves the highest R@1 (0.1675) and MRR (0.2328), while $r = 8$ yields the best R@5 (0.2960). These differences are small, but the pattern is consistent with the hypothesis that lower-rank adapters act as an implicit regularizer when training data is limited. With only 18,000 samples, a more constrained adapter may be less prone to fitting noise in the training captions.

Model	R@1 $\uparrow$	R@5 $\uparrow$	R@10 $\uparrow$	MRR $\uparrow$	Med. Rank $\downarrow$
Vanilla CLIP (zero-shot)	0.1090	0.1935	0.2380	0.1564	110.0
SciCLIP ($r = 4$)	0.1675	0.2920	0.3585	0.2328	36.0
SciCLIP ($r = 8$)	0.1655	0.2960	0.3545	0.2324	34.0
SciCLIP ($r = 16$)	0.1640	0.2930	0.3585	0.2312	34.0

Table 2: Cross-modal retrieval results on the 2,000-candidate evaluation set. All LoRA-adapted variants improve over the zero-shot baseline, though absolute recall values remain moderate.

3. **Diminishing returns at higher rank:** The $r = 16$ configuration does not improve over $r = 8$ on any metric and shows slight degradation on R@1 and MRR. This suggests that under our training budget, increasing adapter capacity beyond $r = 8$ offers no benefit and may lead to mild overfitting.

5 Qualitative Examples

To complement the quantitative evaluation, we inspect selected retrieval results.

- **Query:** “*attention mechanism transformer architecture diagram.*” Vanilla CLIP returned general block diagrams and unrelated line plots, likely matching on surface-level terms like “architecture.” SciCLIP retrieved a transformer encoder-decoder diagram with multi-head attention blocks, suggesting improved alignment between technical vocabulary and visual structure.
- **Query:** “*training loss curve comparison baseline.*” Vanilla CLIP returned bar charts and tables containing the word “comparison.” SciCLIP returned line plots showing training and validation loss trajectories, a more semantically appropriate match.

These examples illustrate that the adapted model captures domain-specific associations that the zero-shot model misses, though these are selected examples and may not be representative.

6 Interactive Retrieval System

We also build a retrieval system for comparing SciCLIP against the vanilla CLIP baseline on arbitrary queries. The system has two stages: pre-indexing and real-time search.

6.1 Pre-indexing Pipeline

We pre-encode all figure images into 512-dimensional vectors using the LoRA-adapted vi-

sion encoder, L2-normalize them, and store them in a FAISS (Johnson et al., 2021) index. For our 2,000-candidate pool we use an exact inner-product index (IndexFlatIP); for larger pools the code falls back to an IVF (Inverted File) index with Voronoi partitioning for sub-linear search.

A separate baseline index is built with the frozen (unadapted) CLIP vision encoder so that both models retrieve from the same candidate set. Indices and figure metadata (paths, captions, arXiv IDs) are saved to disk for fast loading.

6.2 Real-time Search Flow

At query time the input text is tokenized and encoded into a 512-dimensional vector by the LoRA text encoder (and, separately, the frozen baseline encoder). Each vector is matched against its corresponding FAISS index, returning the top- k candidates ranked by cosine similarity. The full pipeline—tokenization, encoding, and search over 2,000 vectors—runs in under 50 ms on CPU, which is fast enough for interactive use.

Figure 3 illustrates the complete retrieval flow.

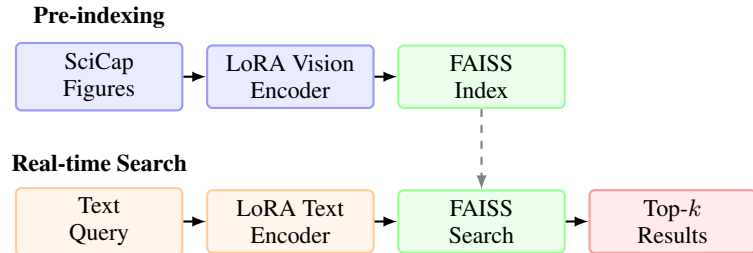


Figure 3: Retrieval system architecture. Pre-indexing: figure embeddings are pre-computed and indexed. Real-time search: a text query is encoded and matched against the index in real time.

6.3 Gradio Interface

We wrap the retrieval pipeline in a Gradio web application. Results from the LoRA-adapted and baseline models are shown in separate tabs, each displaying the top- k figures as thumbnails with

captions and similarity scores. A set of example queries is provided (e.g., “*attention mechanism transformer architecture*”) and users can adjust k between 1 and 10 with a slider.

6.4 Computational Cost

Table 3 summarizes the computational costs of each system component.

Component	Time	Hardware
Image encoding (2k figs)	~45 s	A100 GPU
FAISS index build	<1 s	CPU
Text encoding (per query)	~15 ms	CPU
FAISS search (per query)	<1 ms	CPU
End-to-end query latency	~50 ms	CPU

Table 3: Computational cost of system components. Image encoding is a one-time offline cost; query-time latency is dominated by text encoding.

7 Limitations

Several limitations should be noted. First, our training and evaluation sets are drawn from the same 20,000-sample subset of SciCap; we do not evaluate on a separately held-out test set, which limits claims about generalization. Second, the absolute retrieval accuracy remains low—an R@1 below 17% means the correct figure is the top result less than one-fifth of the time. This reflects both the inherent difficulty of the task (many scientific figures are visually similar) and the limited scale of our training data. Third, we evaluate only a single backbone (clip-vit-base-patch32); larger or more recent vision-language models may yield different conclusions about the effectiveness of LoRA adaptation. Finally, our rank ablation covers only three values; a finer-grained sweep and multiple random seeds would strengthen the analysis.

8 Conclusion

We presented SciCLIP, a LoRA-based adaptation of CLIP for scientific figure retrieval. With approximately 0.5% trainable parameters, the method reduces the Median Rank of correct retrievals from 110 to 34 and improves MRR from 0.1564 to 0.2324 on a 2,000-candidate evaluation set. An ablation over LoRA ranks suggests that smaller ranks ($r = 4, 8$) generalize better than $r = 16$ under limited training data, consistent with the view

that low-rank constraints provide implicit regularization. The moderate absolute performance suggests that LoRA alone is not sufficient to close the domain gap, and points to the need for larger-scale training data, stronger backbones, and possibly hybrid retrieval approaches. In future work, we plan to scale training to the full SciCap dataset (approximately 330,000 samples) and explore incorporating OCR-extracted text from within figures to capture fine-grained textual information that image encoders alone may miss.

References

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A Hyperparameter Details

Table 4 lists the hyperparameters used in our experiments.

Hyperparameter	Value
Base Model	openai/clip-vit-base-patch32
Optimizer	AdamW
Learning Rate	1×10^{-4}
Weight Decay	0.01
Batch Size	64
Epochs	5
LR Scheduler	Cosine Annealing
LoRA Target Modules	q_proj, k_proj, v_proj
LoRA Alpha (α)	16
LoRA Dropout	0.1

Table 4: Training hyperparameters.